

# Structure-Adaptive Spectral Protocol (SASP)

## A Rigorous Reformulation: Spectral Fingerprints as Conjecture Generators, Not Theorem Verdicts

— A critical rewriting of v1.0, in which all "theorems" are demoted to observations and conjectures

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**Abstract.** This paper is a critical rewriting of SASP v1.0. v1.0 proposed a four-stage data-driven pipeline (sampling → arithmetic alignment → spectral fingerprinting → spectral matching) intended to convert "tool-guessing" in mathematical research into "fingerprint matching," with the Collatz stopping-time sequence  $\sigma(n)$  as a case study, and claimed to have obtained three "theorems" and to have "closed" the Markov spectral-gap approach. This revised edition retains the pipeline as a legitimate **experimental-mathematics conjecture generator**, but makes six fundamental corrections: (1) it introduces a **fingerprint-strength hierarchy** — distinguishing "theorem-level fingerprints" (characterization theorems, where a match is a verdict) from "statistical-level fingerprints" (Hurst exponent, spectral energy fractions, Walsh–Hadamard support; conjecture-only); (2) it demotes all "proved theorems" of v1.0 to **numerical observations and testable conjectures**; (3) it establishes a **sub-proposition delineation principle** — any "applicable/not applicable" claim must specify the sub-proposition in question (pointwise convergence / stopping-time distribution / definability / p-adic structure); (4) it **withdraws** the claim of "closing the Markov spectral-gap approach," which would require an independent operator-level argument that v1.0 did not provide; (5) it removes or flags every reference that cannot be publicly verified; and (6) it makes explicit the requirements of **control experiments and falsifiability**. The revised conclusion: SASP is a **conjecture generator**, not a theorem-discovery machine. It closes no research direction, but it does provide a list of testable conjectures and a methodological cautionary note for subsequent rigorous work. Based on the structural ceiling argument "distribution → pointwise"

(§6), this edition further notes: for the Markov spectral-gap line, stitching other tools is a *necessary* condition for solving the full problem (argument-level); for the full problem itself, stitching is the most likely resolution form (judgment-level).

## o. Revision Note

This revised edition corrects the following defects of v1.0, item by item. Each correction is developed further in the body of the paper.

| #  | Defect in v1.0                                                                                                | Treatment in this revised edition                                                                                       |
|----|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| R1 | Numerical observations on finite samples were stated as "proved theorems" (P3/P6/P7)                          | All demoted to "observations" and "conjectures," with evidence levels made explicit (§3, §4)                            |
| R2 | Statistical features (Hurst, spectral energy fractions, WHT support) were used as decisive fingerprints       | A fingerprint-strength hierarchy L1/L2 is introduced (§2.2); statistical features serve as weak hints only              |
| R3 | "Closing the Markov spectral-gap approach" was claimed without delimiting the object or providing an argument | Withdrawn; reformulated as "the question requires an independent argument at the level of sub-proposition B" (§3.5, §6) |
| R4 | Citations to works that cannot be publicly retrieved (Hateley 2026, DeFranco 2026, ACT framework, etc.)       | Removed; only verifiable public references are retained (§7)                                                            |
| R5 | Invented non-standard terminology (Arithmetic Chirality Topology, "modular-4 handedness", ghost cycles)       | Standard phrasing adopted throughout, or explicitly labeled as "working definitions"                                    |
| R6 | No control experiments, no randomized baseline; the method was unfalsifiable                                  | Falsifiability and control-experiment requirements made explicit (design provided; not yet executed, §2.4)              |

## 1. Introduction: The Demand for a Systematic Approach to Tool Selection

In arithmetic dynamics and number theory, choosing the right formal tool (algebraic geometry, analytic number theory, ergodic theory, automata theory, harmonic analysis) has long relied on intuition and analogy. v1.0's observation — that this "tool-guessing" process lacks system and reproducibility — stands, and this revised edition retains that motivation.

The revision must, however, make an epistemological fact explicit: **tool selection operates at two distinct levels of strength that must not be conflated.**

- **Conjecture generation (heuristic):** numerical exploration suggests "this problem may exhibit structure X." This is the legitimate function of experimental mathematics; its output is a testable conjecture.
- **Theorem verdict:** confirming "this problem is structure X" or "method M does not apply to problem P." This requires characterization theorems, proofs, or decisive counterexamples. No numerical observation can support it.

The fundamental error of v1.0 was to treat the former as the latter. Everything in this revised edition unfolds under this distinction.

## 2. Methodological Framework (Revised)

### 2.1 The Pipeline: Four Stages (retained from v1.0)

1. **Sampling and observation:** take the first  $N$  terms of a sequence  $f(n)$ , with fixed random seeds for reproducibility.
2. **Arithmetic alignment:** reparameterize by  $p$ -adic valuations, modular residue classes, and branch structure, stripping known trivial layers.
3. **Spectral fingerprinting:** apply DFT, Walsh–Hadamard, wavelet, and Hurst analysis layer by layer.
4. **Spectral matching:** compare fingerprints against known structures — **producing conjectures, not verdicts.**

### 2.2 NEW: Fingerprint Strength Hierarchy (core of this revision)

Not all "fingerprints" carry equal decisiveness. Two levels are distinguished:

| Level                   | Type of fingerprint                                      | Examples                                                                                                                                                                    | Force of a match                                                |
|-------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| L1<br>Theorem-level     | Deciding conditions defined by characterization theorems | Christol's theorem ( $p$ -automatic $\Leftrightarrow$ algebraic generating function over $F_p$ ), Cobham's theorem, Mahler's 2-kernel, spectral-measure type classification | Match is a verdict (equivalence relation; falsifiable)          |
| L2<br>Statistical-level | Numerical statistical features                           | Hurst exponent, spectral energy fractions, WHT nonzero-coefficient                                                                                                          | Conjecture only (weak hint; no verdictive force; holds for many |

| Level | Type of fingerprint | Examples                           | Force of a match      |
|-------|---------------------|------------------------------------|-----------------------|
|       |                     | ratio, autocorrelation decay shape | unrelated structures) |

**Correction (R2):** v1.0 used L2 statistical features to make L1-level "proved" claims. This edition stipulates that L2 observations are always stated as "observation  $O_k$ ", never as "theorem." An L1 verdict may be claimed only when the corresponding characterization theorem genuinely applies to the object in question.

## 2.3 NEW: Sub-proposition Delineation Principle

"The Collatz problem" is not a single proposition but a family of propositions. Any claim about the applicability of a method must specify the sub-proposition at issue:

| Sub-proposition                 | Content                                                              | Typical tools                                                                   |
|---------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| A · Pointwise convergence       | $\forall n$ : the orbit eventually reaches 1 (the conjecture proper) | Pointwise / analytic methods                                                    |
| B · Stopping-time distribution  | Statistical behavior of $\sigma(n)$ ; typical orbit behavior         | Stochastic models, ergodic theory, spectral methods (Kontorovich–Lagarias, Tao) |
| C · Definability                | Whether $\sigma \bmod 2$ is automatic / Presburger-definable         | Automata theory (Allouche–Shallit)                                              |
| D · p-adic / spectral structure | $v_2$ -fractal spectra, mod- $2^k$ structure                         | Harmonic analysis, p-adic analysis                                              |

**Correction (R3):** v1.0 claimed to "close the Markov spectral-gap approach" without specifying the sub-proposition. This edition stipulates that any conclusion about applicability must be written as "for sub-proposition X, the applicability of method M requires argument Y."

## 2.4 NEW: Falsifiability and Control-Experiment Requirements

An effective exploratory pipeline must be able to fail. Two requirements are mandatory:

- **Randomized baseline:** run the same fingerprint extraction on equally long pseudo-random sequences (or permuted versions of the original sequence) as a null hypothesis. An observation  $O$  is worth reporting only if it deviates significantly from the baseline.
- **Falsification condition:** each conjecture must state "which fingerprint pattern, if observed, would count as refuting this conjecture." If a matching

process always succeeds, its output carries no information.

**Status:** the control experiments above are **not yet executed** in this revised edition. Accordingly, every observation in §3 is labeled "value as reported in v1.0; not independently reproduced; not compared against a randomized baseline." This is an honest statement of data status, not a claim of results.

### 3. Application: The Collatz Stopping-Time Sequence — Data Report (observations, not theorems)

Everything below is either data reported by v1.0 or an elementary, directly verifiable fact. This edition adds no new experimental numbers.

#### 3.1 Observation O1: 2-adic recursion stratification (directly verifiable)

For the total stopping time  $\sigma(n)$  (of the Collatz map  $T$ ), the following recursions are elementary and verifiable:

$$\sigma(2m) = 1 + \sigma(m), \quad \sigma(n) = v_2(n) + \sigma(\text{odd-core}(n))$$

This legitimizes a 2-adic stratification (sub-proposition D): after stripping the trivial  $v_2$  layer, the core signal is  $\sigma$  on the odd core.

#### 3.2 Observation O2: Exact DFT structure of $v_2(3n+1)$ (independently verifiable)

v1.0 (S-E2) reported that the DFT of  $v_2(3n+1)$  over  $N = 2^K$  has an exact dyadic-fractal structure:

$$G[k] = \sum_{i=1..K} 1[(N/2^i) \mid k] \cdot (N/2^i) \cdot \omega^{k \cdot e_i} + \omega^{k \cdot e_{\text{base}}}, \quad e_i = (2 \cdot 4^{i-1} - 2) / 3$$

This formula follows by elementary recursion from  $v_2$  and constitutes an **independently verifiable computational fact** (L1-level computation, but computation only; it is not a proposition about the Collatz conjecture). Its connection to carry dynamics (see Conjecture C2) has not been rigorously proved.

#### 3.3 Observation O3: Hurst exponent $H \approx 0.608$ (weak evidence)

v1.0 (S-F5) reported a Hurst estimate  $H \approx 0.608 > 0.5$  for the detrended  $\sigma$  residual. This edition notes:

- the estimate is based on a single sample with  $N \leq 20000$ , with no confidence interval, window parameters, or estimation method (DFA / R-S / periodogram) reported;
- $H > 0.5$  can arise from banal causes such as non-stationarity, residual trend, and finite-sample bias;
- no randomized baseline was used.

**Verdict:** O3 can at most generate a conjecture (see C1); it constitutes no evidence of "long-range dependence" or a "non-mixing subspace."

### 3.4 Observation O4: Full WHT support of $\sigma \bmod 2$ (no discriminative power)

v1.0 (S-J3) reported that  $\sigma \bmod 2$  over  $N = 16384$  has 16383/16384 nonzero Walsh–Hadamard coefficients. This edition notes:

- pseudo-random sequences also exhibit full WHT support ( $\approx 100\%$  nonzero);
- while a theoretical link between O4 and non-automaticity exists (the WHT structure of automatic sequences has characterizations), full support itself has **no discriminative power** between "random", "non-automatic", and other complex structures;
- v1.0 itself conceded in P5 that "full support is neither necessary nor sufficient"; this edition carries that concession through: O4 supports no testable conclusion.

### 3.5 Observation O5: Determinism and spectrum $\{1 \times 2, 0 \times 14\}$ of the mod-16 transition (reproducible computation)

That  $\bmod 2^k$  determines the first  $k$  parity steps is a classical fact since Terras (sub-proposition D). The mod-16 transition spectrum  $\{1 \times 2, 0 \times 14\}$  reported by v1.0 (F7/D25) is a **reproducible finite computation**, accepted by this edition as a computational fact.

**Important withdrawal (R3):** v1.0 inferred from this that "the Markov spectral-gap approach is closed." This edition withdraws that claim, for three reasons: (1) the spectral-gap method studies the distributional evolution of random initial values and weighted/infinite-dimensional transfer operators (sub-proposition B), not a finite mod-16 matrix; (2) the determinism of a finite matrix does not undermine the legitimacy of statistical models — just as "each individual's height is fixed" and "heights in a population form a distribution" are not in conflict; (3) "closing" would require an operator-level argument (e.g., proving that a necessary condition for the spectral-gap method fails systematically on Collatz-related objects), which v1.0 did not provide. This edition marks "applicability of the Markov spectral-gap method to sub-proposition B" as an **open question**.

## 4. Testable Conjectures from the Observations

The conjectures below are the full positive output of the SASP pipeline in this case study. **None of them is proved**, but each carries an explicit falsification condition.

| ID | Conjecture                                                                                                                                   | Source observation                                      | Falsification condition                                                                                                                                            |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1 | The detrended $\sigma$ residual may exhibit non-exponential correlation decay in some weighted space for the Syracuse transfer operator      | O3                                                      | H falls into a 0.5 neighborhood under larger samples ( $N \geq 10^6$ ) and rigorous DFA/spectral estimation, or an explicit construction explains all observations |
| C2 | A provable connection exists between the DFT fractal exponents $e_i$ of $v_2(3n+1)$ and characteristic periods of carry dynamics             | O2                                                      | The spectrum cannot be constructed within a finite-automaton/carry framework                                                                                       |
| C3 | The broadband spectrum of $\sigma(n)$ may be related to parity imbalance in mod-4 classes (spectrum becomes sparse after "de-chiralization") | v1.0's $S_3$ observation (not independently reproduced) | No precise definition of "de-chiralization" can be given, or sparsity fails under the definition                                                                   |
| C4 | Applicability of the Markov spectral-gap method to sub-proposition B: open question                                                          | —                                                       | An operator-level proof exists (in either direction)                                                                                                               |

## 5. Methodological Audit of v1.0 (self-audit)

This edition records the six methodological defects that invalidated v1.0, as a reusable review checklist:

- 1. Inductive leap:** numerical fingerprints on finite samples ( $N \leq 20000$ ) were written directly as theorems about infinite objects. Numerical evidence can generate conjectures; it cannot close a proof.
- 2. Insufficient discriminative power:** the statistical fingerprints used produce the same output on pseudo-random sequences, with no control group. Collatz cannot be distinguished from "arbitrarily complex sequences."



3. **Unfalsifiability:** match success → "framework applies"; match failure → "candidate for a new structure." The pipeline never fails, so its output carries no information.
4. **Object mismatch:** the determinism of a finite mod-16 matrix was used to "close" stochastic-model methods whose object is infinite-dimensional operators/distributional evolution — a straw-man argument.
5. **Citations and terminology:** references to unverifiable "in preparation" works, and invented non-standard terminology, undermined verifiability.
6. **Level conflation:** L2 statistical features were treated as L1 verdicts — the common root of all defects above.

## 6. Resolution-Form Outlook: The Stitching Argument and Strategic Conclusions (NEW)

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This section advances the methodological conclusions of the paper to the level of "resolution form." It explicitly distinguishes two strength levels: **argument-level conclusions** (derived from the structural ceiling) and **judgment-level conclusions** (methodological predictions, not theorems).

### 6.1 Position Statement: SASP Is a Methodology, Not a Theorem Machine

The deliverable of SASP is the **methodological framework itself** — a reproducible exploratory pipeline (sampling → alignment → fingerprinting → matching). Its value lies in the reliability of the process and the testability of its outputs, not in any single finding. Any use that claims "theorem discovery" or "closing a research line" exceeds its capacity. The error of v1.0 was to misuse a methodology as a theorem machine; every correction in this revised edition revolves around that misuse.

### 6.2 The Structural-Ceiling Argument (argument-level)

For the Markov spectral-gap / stochastic-model line, the following argument holds:

Premise 1: distributional methods (spectral gap, stochastic models, ergodic theory) produce measure-theoretic conclusions and in principle cannot be upgraded to pointwise universal propositions;  
Premise 2: the Collatz conjecture proper (sub-proposition A) is a pointwise universal proposition ( $\forall n$  the orbit reaches 1);  
Corollary 1: no single distributional method can fully solve the Collatz conjecture;  
Corollary 2: for the Markov spectral-gap line, if the full



problem is solved, stitching (combining other tools) is a necessary condition.

This argument does not depend on any numerical experiment; its force comes from the "distribution  $\rightarrow$  pointwise" nature of the tools. It must be stressed: Corollaries 1–2 are **argument-level conclusions**, conditional on the premises above.

### 6.3 The Stitching-Puzzle View (schematic)

If Collatz is resolved by stitching, its structure can be schematized as several puzzle blocks (**a methodological schematic, not a theorem**):

- **Block 1 • typical behavior of almost all orbits** (a set of measure 1): the results of Terras, Kontorovich–Lagarias, and Tao are already stitched in here — this is the largest block, and it is where the "Markov spectral-gap line" lives.
- **Block 2 • exceptional-set control / pointwise closure**: the piece that upgrades "almost all" to "all." Currently missing; it is the bottleneck of any stitching solution — the capping tool is yet to be invented.
- **Block 3 • p-adic / structural understanding**:  $v_2$  recursions, mod- $2^k$  structure, etc. (O1, O2 in this case study belong here); partially available, not yet stitched into proof components.
- **Block 4 • definability / negative results**: automaticity, Presburger, method boundaries — exclusionary knowledge, partially settled.

### 6.4 Strategic Conclusion: Do Not Overcommit; Stitch in Parallel

- **For the Markov spectral-gap line: "stitching is necessary" is argument-level** (§6.2); the ceiling of this line lies not in insufficient effort but in the nature of the tools. Yet Block 1 is a large, already-stitched piece, and continuing to expand it (the direction of Tao et al.) is entirely correct.
- **For the full problem: "stitching is most likely" is judgment-level** — logically, a brand-new unified method cannot be excluded (nobody can prove it does not exist), so this section does not claim "it must be a chimera"; it claims only "for the Markov line, stitching is necessary."
- **Action recommendation**: advance all blocks in parallel + make each block's boundary explicit (sub-proposition delineation) + prioritize the search for Block 2 (a pointwise-closure tool). "Do not overcommit" means do not bet everything on a single method solving the whole problem — it does not mean stop researching.

**Relation to v1.0:** v1.0 pointed, by an invalid argument, in the direction of "single methods have a ceiling." This revised edition provides a sound argument and makes the strength levels explicit: argument-level for the Markov line, judgment-level for the full problem. The paper is positioned as a methodology (§6.1); its value lies in the process and the warning, not in any "closure."

## 7. Conclusion

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The revised SASP has a clear position:

- it is a **reproducible exploratory pipeline** whose outputs are lists of conjectures and records of negative paths (the F1–F6 failure records retain their reference value, but likewise constitute no "closure" claims);
- it **produces no theorems** and **closes no research direction**;
- its actual contribution in this case study: two independently verifiable computational facts (O1, O2), a set of testable conjectures (C1–C3), and one explicitly open question (C4);
- its most important contribution is a methodological warning: **fingerprint matching can only answer "in which direction is it worth guessing?", never "is this direction right?"** — the latter always requires characterization theorems, proofs, or decisive counterexamples.

This paper does not solve the Collatz conjecture, nor does it declare any method "inapplicable." It does one thing only: it corrects the errors of v1.0 and leaves a legitimate experimental-mathematics pipeline in its proper place. At the same time, it puts the resolution-form question on the table: for the Markov spectral-gap line, stitching is necessary (argument-level); for the full problem, stitching is the most likely form (judgment-level) — a single line capping the whole problem lies outside the picture of this paper.

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**References** (only publicly verifiable works are listed; the v1.0 entries Hateley 2026, DeFranco 2026, Dhiman 2026, and the ACT framework are removed because they could not be publicly retrieved)

Conway, J. H. (1972). Unpredictable iterations. *Proc. 1972 Number Theory Conf.*

Terras, R. (1976). A stopping time problem on the positive integers. *Acta Arith.*

Lagarias, J. C. (1985). The  $3x+1$  problem and its generalizations. *Amer. Math. Monthly.*

Allouche, J.-P., & Shallit, J. (2003). *Automatic Sequences*. Cambridge University Press.

Kontorovich, A., & Lagarias, J. C. (2009). Stochastic models for the  $3x+1$  problem. *arXiv:0910.1944*.

Tao, T. (2019). Almost all orbits of the Collatz map attain almost bounded values. *arXiv:1908.08104*.

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Data-status statement: all numerical values in §3 are taken from the v1.0 report; they have not been independently reproduced or compared against a randomized baseline; the control-experiment design is given in §2.4.  
This revised edition contains no new experimental data.